

DATA-DRIVEN FRAUD MITIGATION **AN AUDITED MACHINE LEARNING APPROACH**

Capstone Project for PG Dip in AI/ML

Presented By

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THE ECONOMIC IMPACT OF FRAUD

\$4.76

LOSS MULTIPLIER

Every \$1 stolen costs \$4.76 in fees,
labor, and recovery.

6.58%

CONCENTRATION

Target anomalies represent a
focused subset of 10,000 logs.

Source: LexisNexis Risk Solutions 2024 True Cost of Fraud Study

THE DECISION **DILEMMA**: RISK VS. FRICTION

THE **FRAUD** THREAT

Unchecked False Negatives (missed fraud) lead to catastrophic direct revenue leakage and regulatory penalties.

- Direct asset depletion
- Compliance violations
- Network insecurity

THE **CUSTOMER** TAX

Aggressive False Positives (false blocks) trigger "Customer Friction," spikes in call volume, and long-term churn.

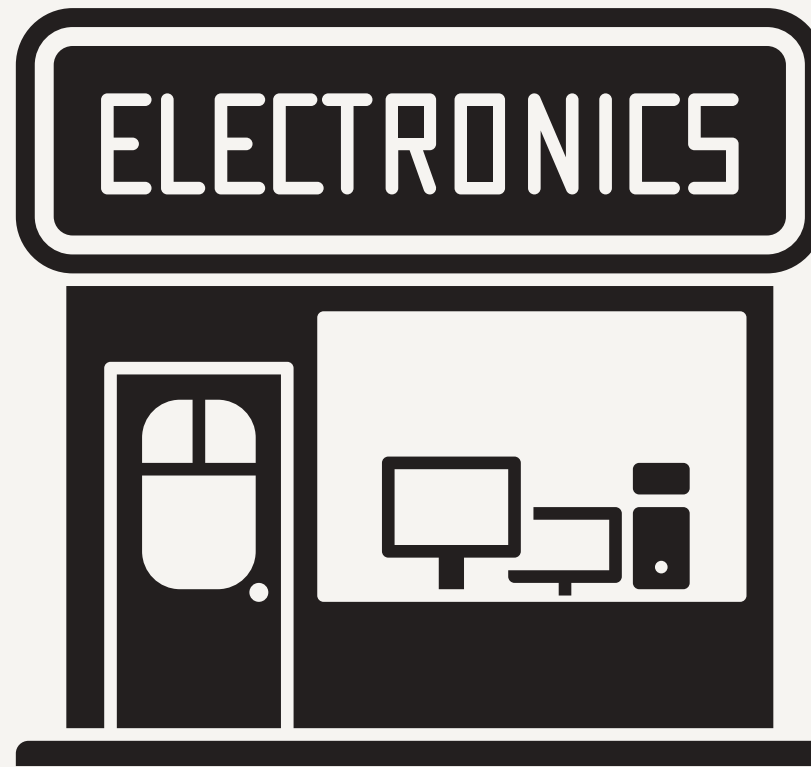
- Brand reputation damage
- Operational overload
- Revenue abandonment

ANATOMY OF IDENTIFIED **RISK** PROFILES



GEO-TEMPORAL

Late-night transactions originating from Florida retail hubs.



HIGH-VALUE RETAIL

Sudden high-ticket purchases in the Electronics sector.



TRAVEL OUTLIERS

Unusually massive expenses registered in the travel segment.

TURNING RAW DATA INTO **STRATEGIC INTELLIGENCE**



Categorical Encoding: Expanding merchant types and locations into high-dimensional binary vectors for algorithmic consumption.



Feature Scaling: Normalizing transaction amounts to prevent extreme value bias and ensure variance stability.



Synthetic Balancing (SMOTE): Counteracting class asymmetry by generating intelligent minority instances to focus the model on fraud signals.



Leakage Prevention: Rigid separation of training and validation sets to ensure "real-world" predictive validity.

PREDICTIVE **MODELING** ARCHITECTURE

Deploying Challenger Models to Optimize Fraud Capture Efficiency

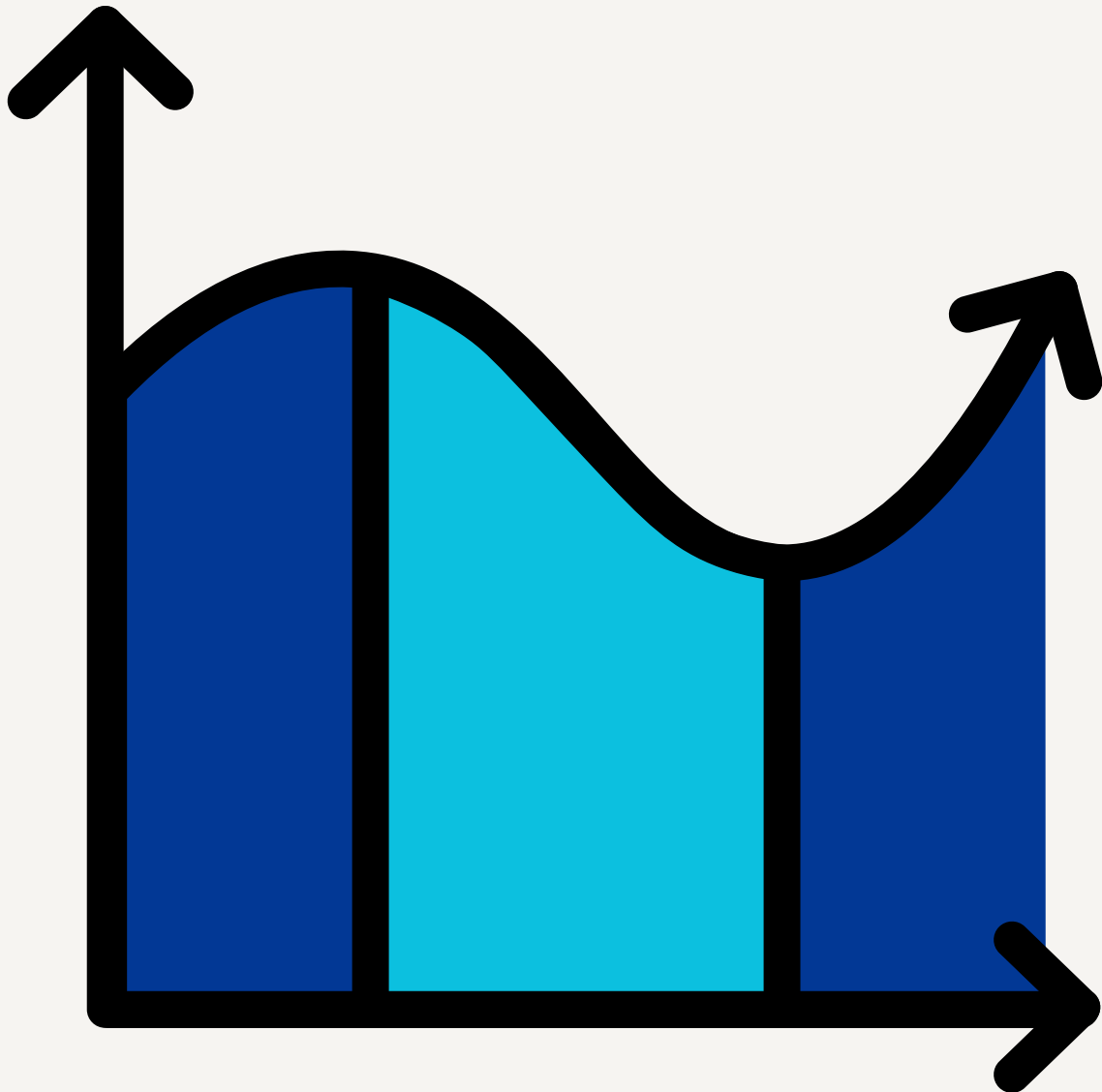
AUPRC OPTIMIZATION

WHY **ACCURACY** FAILS

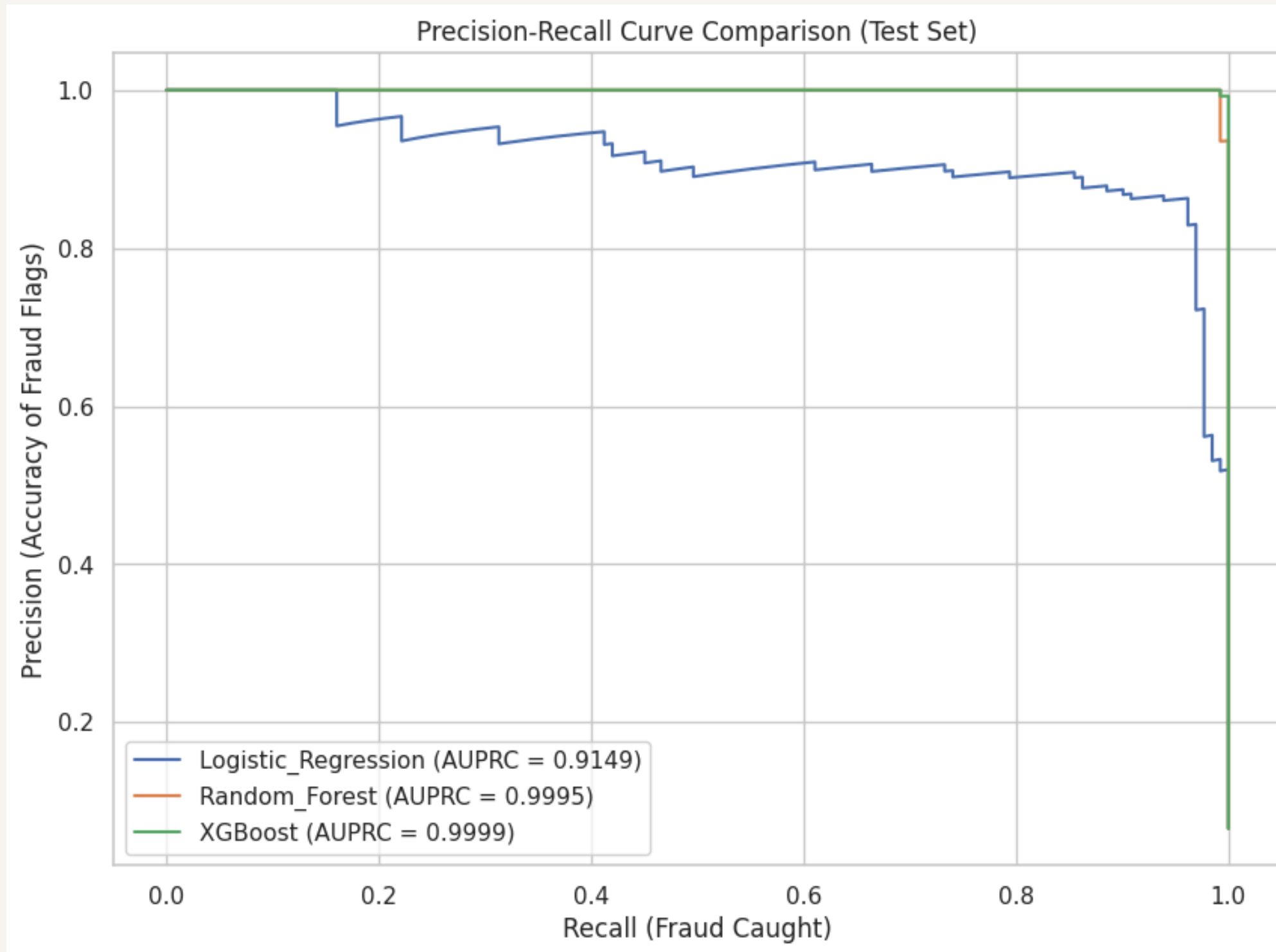
In a network where 93.42% of events are legitimate, a system that predicts "no fraud" achieves high accuracy while protecting zero assets. We optimize for the **Area Under the Precision-Recall Curve (AUPRC)**.

$$\text{AUPRC} = \int_0^1 p(r) dr$$

This mathematically targets the "Sweet Spot" between **aggression** and **conservatism**.



MODEL PERFORMANCE COMPARISON



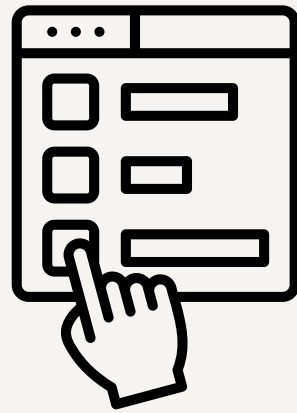
AUPRC

- Logistic Regression: 0.914927
- Random Forest: 0.999508
- XGBoost; 0.999942

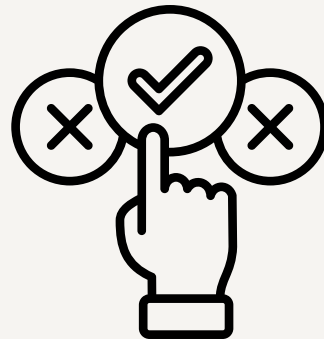
The **Gradient Boosted approach (XGBoost)** demonstrated superior ability to distinguish complex behavioral fraud rules with 99.99% efficiency.

TRUSTING THE MACHINE

RISK ATTRIBUTION



Feature Weights: The model identified 'Amount' and 'Location_FL' as the primary drivers of anomaly detection.

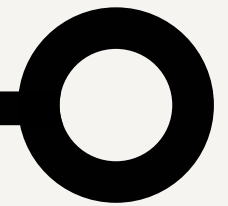
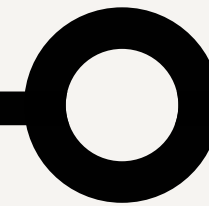
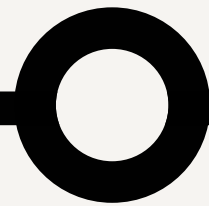
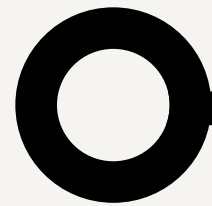
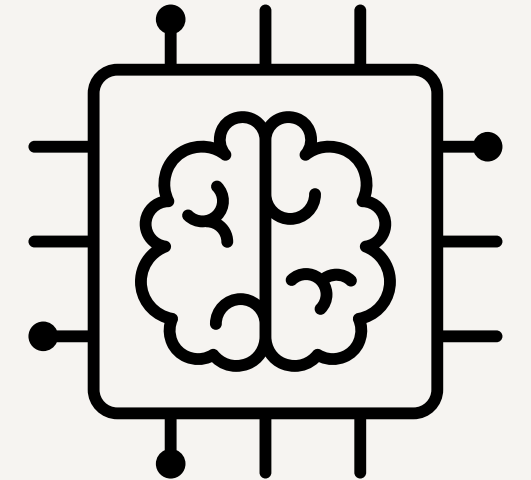
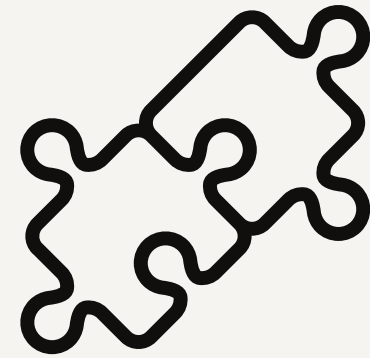
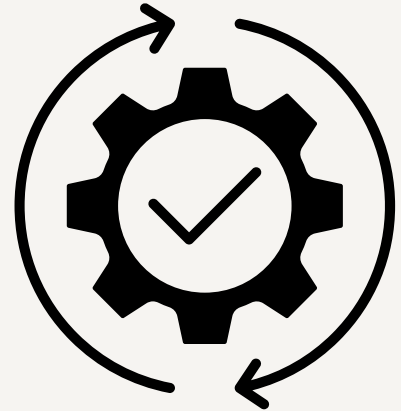


Decision Clarity: Logic is traceable, showing high-risk flagging for travel outliers above the 95th percentile.



Audit Readiness: The framework supports regulatory scrutiny by documenting exactly why a transaction was declined.

DEPLOYMENT ROADMAP



PILOT PHASE

Shadow mode testing on live gateway streams.

INTEGRATION

Hard-blocking enabled for high-confidence alerts.

OPTIMIZATION

Feedback loop from customer support for alert tuning.

FULL ROLLOUT

Autonomous mitigation across all retail channels.

SECURING THE FUTURE

Our machine learning framework transforms reactive loss centers into proactive value drivers. By detecting the invisible rules of fraud, we protect billions in assets while ensuring a seamless experience for our most loyal customers.



"The strongest ROI comes from preventing the disaster before it arrives."

